

Supplementary Material

MAML-Select: An Online Adaptive Client Selection Method for Federated Learning via Meta-Learning

Concise revision supplement

S1. Experimental Protocol

This supplement records the additional evidence used to support the revised letter. The focus is deliberately compact: reproducibility settings, MAML-Select resource savings, statistical support, sensitivity and ablation evidence, and selector-scaling behaviour. The complete baseline table is retained in the main manuscript; here we highlight the results most directly connected to the reviewers’ requests.

Table 1: Shared experimental configuration.

Component	Setting
Clients / selected clients	$N = 100$, $K = 10$
Communication rounds	200 for Fashion-MNIST/CIFAR-10; 150 for CIFAR-100 analysis
Local training	$E = 5$ epochs, batch size 32, client LR 0.01
Data heterogeneity	Dirichlet non-IID split, $\alpha = 0.5$
Models	LightCNN-9 for Fashion-MNIST; ResNet-18 for CIFAR-10/100
MAML-Select policy	MLP 6–64–64–1 with 4,673 parameters
Meta optimization	One inner look-ahead step, $\beta = 0.01$; Adam outer update, $\eta = 0.001$
Default trade-off weight	$\lambda = 0.5$
Seeds	42, 123, 2026

The heterogeneous-client model uses three device tiers with fixed seeded assignment: low-end clients (20%), mid-range clients (50%), and high-end clients (30%). Per-round wall-clock time is straggler-bound by the slowest selected client. Energy is accumulated from the selected clients’ tier power and modeled training time. Since all methods select $K = 10$ clients per round, MAML-Select’s resource improvements come from adaptive client choice rather than from reducing the number of participating clients.

S2. Resource and Statistical Evidence

Table 2 summarizes MAML-Select relative to FedAvg on the shared seeds. Across all three datasets, MAML-Select keeps full participation coverage while reducing cumulative training compute and energy. The savings are largest on Fashion-MNIST and CIFAR-10, and remain measurable on CIFAR-100 despite the harder model and evaluation setting.

Table 2: MAML-Select resource profile relative to FedAvg over shared seeds. Positive reductions indicate lower resource use for MAML-Select.

Dataset	Acc. (%)	TFLOPs ↓	Energy ↓	Coverage
Fashion-MNIST	90.11 ± 0.47	12.8%	16.4%	100%
CIFAR-10	75.63 ± 1.43	21.5%	24.7%	100%
CIFAR-100	58.15 ± 0.51	1.7%	4.7%	100%

Table 3: Paired statistical support for MAML-Select resource reductions versus FedAvg. Reported p values are two-sided paired t -tests over shared seeds; $|d_z|$ is the standardized paired effect size.

Dataset	Metric	Mean reduction	$ d_z $	p
Fashion-MNIST	TFLOPs	17.93	5.93	0.009
	Energy (Wh)	943.58	6.74	0.007
CIFAR-10	TFLOPs	1792.16	12.68	0.002
	Energy (Wh)	1187.44	18.11	0.001
CIFAR-100	TFLOPs	107.80	5.15	0.012
	Energy (Wh)	169.65	10.89	0.003

Tier-level participation. Table 4 reports, for each method, the fraction of selections that fall in each device tier, together with client coverage. The device pool is 20% Tier 1 (slow, 1.0 $\times$), 50% Tier 2 (medium, 2.0 $\times$), and 30% Tier 3 (fast, 4.0 $\times$), so a tier-proportional selector reads 0.20/0.50/0.30. Two patterns stand out. The purely system-aware

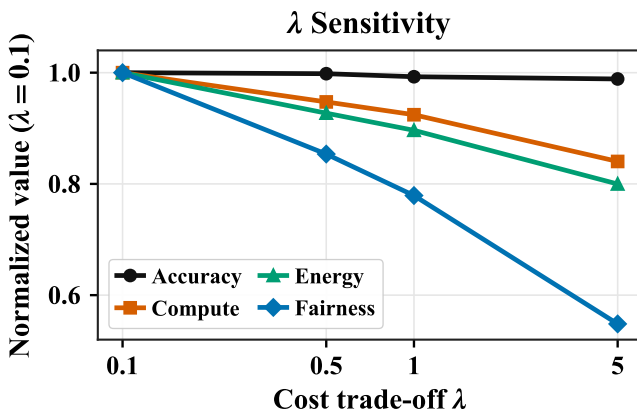
selectors concentrate on the fast tier and abandon the slow tier: FedCS selects *no* Tier 1 client (0.00) and sends 80 to 83% of its selections to Tier 3, at only 10% coverage. MAML-Select instead shifts selection toward faster tiers to lower latency and energy (Tier 3 rises to 0.44 to 0.49), yet it keeps the slow tier represented (0.09) and reaches 100% coverage. This is the tier-level view of the Jain index and coverage results: MAML-Select is cost-aware but never permanently excludes a tier, unlike the deadline- or utility-constrained baselines.

Table 4: Tier-level selection rates (fraction of selections per device tier) and client coverage, mean over seeds. Tier 1 slow / Tier 2 medium / Tier 3 fast; pool proportions 0.20/0.50/0.30.

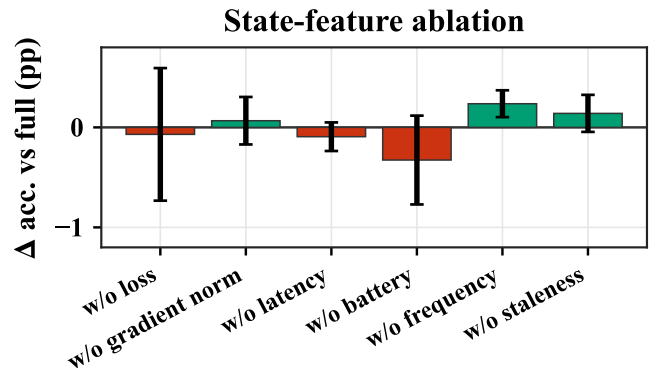
Method	Tier 1	Tier 2	Tier 3	Cov. (%)
<i>Fashion-MNIST</i>				
FedAvg	0.20	0.50	0.31	100
FedCS	0.00	0.20	0.80	10
Oort	0.20	0.37	0.43	10
TiFL	0.33	0.33	0.33	12
FedCor	0.16	0.44	0.41	49
CriticalFL	0.20	0.47	0.33	33
FedGCS	0.20	0.50	0.30	100
MAML-Select	0.09	0.47	0.44	100
<i>CIFAR-10</i>				
FedAvg	0.20	0.50	0.31	100
FedCS	0.00	0.17	0.83	10
Oort	0.20	0.37	0.43	10
TiFL	0.33	0.33	0.33	12
FedCor	0.18	0.56	0.25	31
CriticalFL	0.20	0.48	0.32	46
FedGCS	0.22	0.47	0.31	100
MAML-Select	0.09	0.41	0.49	100

S3. Sensitivity and State-Feature Ablation

The sensitivity study evaluates the cost trade-off weight $\lambda \in \{0.1, 0.5, 1.0, 5.0\}$ on Fashion-MNIST. The selected value $\lambda = 0.5$ gives a balanced operating point: accuracy remains close to the best observed value while compute and energy are reduced relative to the low-penalty setting. The ablation study verifies that the six-dimensional state vector gives stable behaviour; removing one feature at a time changes accuracy only modestly, which indicates that the learned selector is not dependent on a single fragile signal.



(a) λ sensitivity.



(b) State-feature ablation.

Figure 1: Design evidence for MAML-Select. The λ sweep shows the accuracy–resource trade-off, and the ablation plot shows that the policy remains stable under single-feature removals.

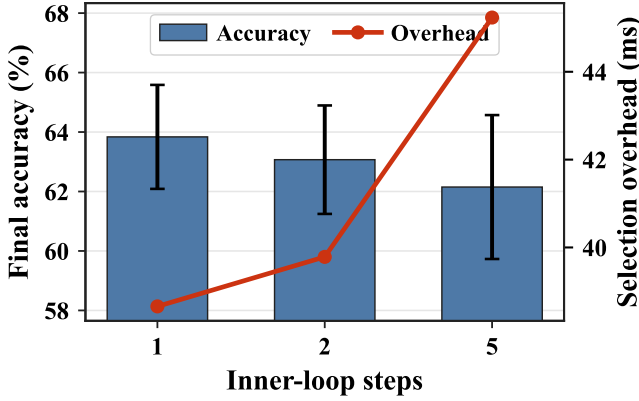
The revised paper adds two further ablations on the selector itself. The inner-step ablation varies the number of inner adaptation steps on CIFAR-10. A single step gives the highest accuracy and the lowest selection overhead, which justifies the one-step inner loop. The selector-width ablation varies the hidden width $h \in \{16, 32, 64, 128\}$ of the 6- h -1 policy on Fashion-MNIST. Accuracy stays within about 0.6 points across this 44 $\times$ parameter range, so the compact

Table 5: Fashion-MNIST sensitivity and ablation summaries.

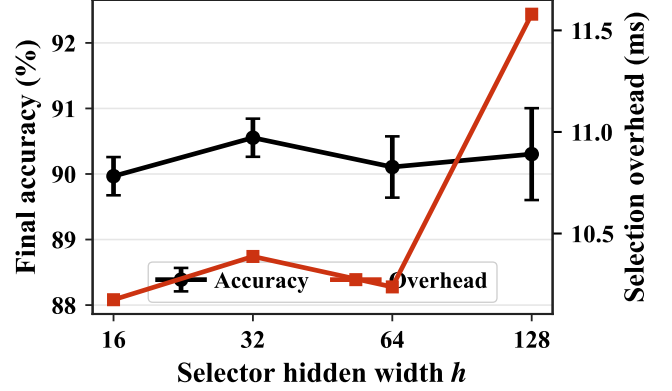
λ	Acc. (%)	TFLOPs	Energy (Wh)
0.1	90.40 ± 0.22	128.3	5176
0.5	90.23 ± 0.55	121.6	4799
1.0	89.74 ± 0.79	118.6	4639
5.0	89.37 ± 1.02	107.8	4140

Variant	Acc. (%)	TFLOPs
Full state	90.23 ± 0.55	121.6
w/o loss	90.16 ± 0.66	122.8
w/o grad. norm	90.30 ± 0.24	122.4
w/o latency	90.14 ± 0.14	118.9
w/o battery	89.90 ± 0.44	122.2
w/o frequency	90.47 ± 0.13	115.0
w/o staleness	90.37 ± 0.19	122.7

$h = 64$ default is robust.



(a) Inner-loop steps (CIFAR-10).



(b) Selector width (Fashion-MNIST).

Figure 2: Selector design ablations. (a) One inner step is the most accurate and lowest-overhead setting. (b) Accuracy is flat across selector width while overhead grows, so the compact $h = 64$ default is sufficient.

S4. Selector Scaling

The selector-scaling probe runs MAML-Select for 10 rounds while increasing $N = \{20, 40, 80, 100, 200, 500, 1000\}$ and keeping $K/N = 0.1$. The measured per-round selection overhead stays nearly flat, between about 20 and 26 ms, even as N grows fifty-fold to 1000, and is small compared with local training time. This supports the complexity argument in the main paper: scoring all available clients with a fixed-size MLP is linear in the client pool, and at these pool sizes that $O(NP)$ cost is dominated by fixed per-round overhead, so the measured time barely changes.

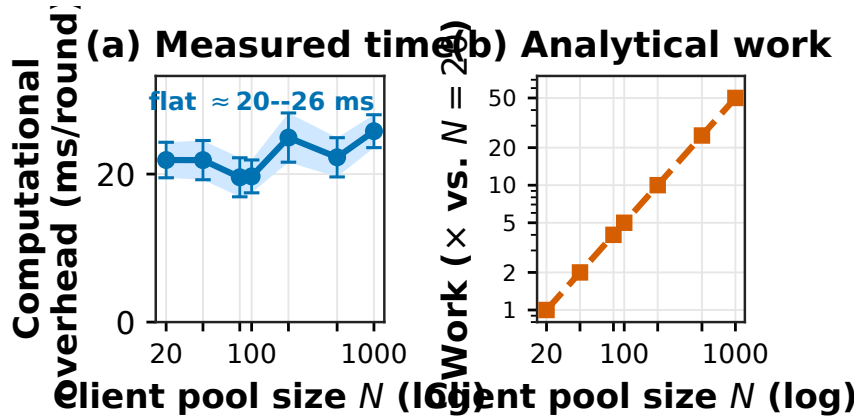


Figure 3: MAML-Select client-selection overhead as the pool size increases, with the selection rate held at 10% ($K/N = 0.1$). Panel (b) plots the analytical operation count $C(N, K) = NP + N \log K + 2KP$ ($P = 4,673$), normalized to its value at $N = 20$.

Table 6: Mean MAML-Select selection overhead in seconds, measured over 10 rounds, for client-pool sizes up to $N = 1000$ with the selection rate held at $K/N = 0.1$.

Method	$N = 20$	$N = 40$	$N = 80$	$N = 100$	$N = 200$	$N = 500$	$N = 1000$
MAML-Select	0.02190	0.02189	0.01957	0.01969	0.02493	0.02227	0.02580

S5. Extended Algorithm

Algorithm 1 gives the full MAML-Select procedure, expanding Algorithm 1 of the main paper with the state-feature construction, the cold-start rule, the exploration step, and the first-order inner and outer updates. The meta-policy h_ϕ is the 6-64-64-1 MLP. No client raw data leaves the device: the server observes only the six-dimensional state vectors and scalar cost feedback. The procedure is the exact specification used to produce all reported results, so the experiments are reproducible from these steps and the stated hyperparameters.

Algorithm 1 MAML-Select (extended)

Require: clients $\{1, \dots, N\}$; rounds T ; cohort K ; inner LR β ; outer LR η ; trade-off λ ; target latency T_{target} ; exploration size ϵ

- 1: Initialize global model θ_1 , meta-policy ϕ_1 , and the Adam state for ϕ
- 2: $\mathcal{P} \leftarrow \emptyset$ ▷ pending feedback from the previous round
- 3: **for** $t = 1, \dots, T$ **do**
- 4: $\mathcal{A}_t \leftarrow$ clients available this round
- 5: **for** $i \in \mathcal{A}_t$ **do** ▷ state features
- 6: $\mathbf{s}_{i,t} \leftarrow [\hat{L}_i, \|\nabla F_i\|, \hat{T}_i, b_i, q_i, \tau_i]$ ▷ loss, grad-norm, latency, battery, frequency, staleness
- 7: **end for**
- 8: **if** $\mathcal{P} \neq \emptyset$ **then** ▷ inner adaptation and first-order outer update
- 9: $\mathcal{D}_{\text{sup}} \leftarrow \{(\mathbf{s}_{j,t-1}, c_{j,t-1}) : j \in \mathcal{P}\}$
- 10: $\phi'_t \leftarrow \phi_t - \beta \nabla_\phi \mathcal{L}(h_\phi; \mathcal{D}_{\text{sup}})$ ▷ one inner step
- 11: $\phi_{t+1} \leftarrow \phi_t - \eta \nabla_{\phi'_t} \mathcal{L}(h_{\phi'_t}; \mathcal{D}_{\text{sup}})$ ▷ first-order meta-update
- 12: **else**
- 13: $\phi'_t \leftarrow \phi_t$; $\phi_{t+1} \leftarrow \phi_t$
- 14: **end if**
- 15: $\hat{c}_i \leftarrow h_{\phi'_t}(\mathbf{s}_{i,t})$ for all $i \in \mathcal{A}_t$ ▷ score candidates
- 16: **if** $t \leq \lceil N/K \rceil$ **then**
- 17: $\mathcal{S}_t \leftarrow$ next K clients in a fixed coverage order ▷ cold start
- 18: **else**
- 19: $\mathcal{S}_t \leftarrow$ Top- K of $(-\hat{c})$, with ϵ random exploratory clients
- 20: **end if**
- 21: Train selected clients; aggregate $\theta_{t+1} \leftarrow \sum_{i \in \mathcal{S}_t} \frac{n_i}{\sum_{j \in \mathcal{S}_t} n_j} \theta_i$ ▷ FedAvg
- 22: **for** $i \in \mathcal{S}_t$ **do** ▷ record cost feedback
- 23: $\rho_i \leftarrow \max(0, T_i - T_{\text{target}})/T_{\text{target}}$ ▷ normalized latency penalty; T_i is realized latency
- 24: $c_{i,t} \leftarrow \lambda \rho_i - \Delta_i$ ▷ Δ_i : local loss reduction, as in Eq. (4)
- 25: **end for**
- 26: $\mathcal{P} \leftarrow \mathcal{S}_t$
- 27: **end for**
- 28: **return** θ_{T+1}

S6. Selector Convergence

Lemmas 1 and 2 of the main paper give the inner and outer updates a one-step descent property. Here we draw the standard consequence for the meta-iterates and then verify both properties empirically over training.

Corollary (approximate stationarity). Suppose the query objective is shared across rounds, $q_t \equiv q$ with q bounded below by q^* , and Lemma 2 holds with step size $0 < \eta \leq 1/L_q$. Summing the per-round descent over $t = 1, \dots, T$ and telescoping gives

$$\min_{1 \leq t \leq T} \|\nabla q(\phi_t)\|_2^2 \leq \frac{2(q(\phi_1) - q^*)}{\eta T} + \frac{1}{T} \sum_{t=1}^T \delta_t^2.$$

The meta-policy therefore reaches an $\mathcal{O}(\beta^2)$ -approximate stationary point at rate $\mathcal{O}(1/(\eta T))$, where the floor is the first-order error $\delta_t^2 \leq L_q^2 \beta^2 \|\nabla g_t(\phi_t)\|_2^2$. In MAML-Select the query objective is mildly non-stationary because client utility drifts, so the bound describes the slowly-varying regime; we report the empirical behaviour in the general case next.

Fig. 4 verifies the two descent properties during a full training run on each dataset. Panel (a) plots the inner-step change $g_t(\phi'_t) - g_t(\phi_t)$; it stays at or below zero in every round (100% of adapted rounds on all three datasets), confirming Lemma 1. Panel (b) plots the query objective relative to its first-round value; it decreases and then stabilizes, which is the empirical signature of outer-loop convergence under the non-stationary target. Panel (c) plots the meta-update

magnitude $\|\phi_{t+1} - \phi_t\|_2$; it shrinks and settles, showing that the policy stops moving as it tracks the utility signal. Together with the global accuracy and loss curves of the main paper, these confirm that the adaptive selector and the federated model both converge in practice, even though a closed-form global guarantee under fully adaptive partial participation remains open.

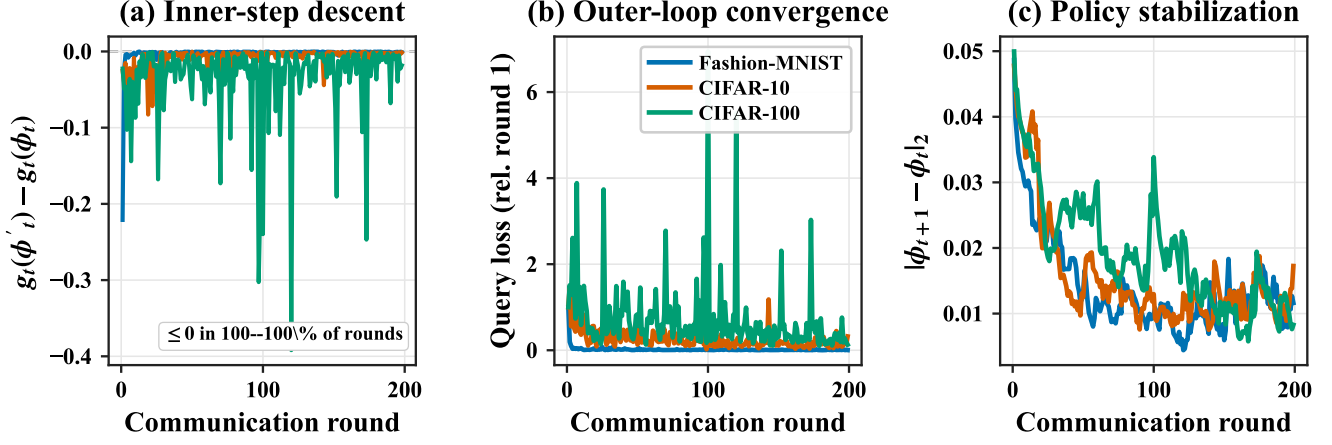


Figure 4: Selector-convergence verification over a full training run. (a) Inner-step support descent $g_t(\phi'_t) - g_t(\phi_t) \leq 0$ (Lemma 1). (b) Query objective relative to round 1, decreasing and stabilizing (outer-loop convergence, Lemma 2). (c) Meta-update magnitude $\|\phi_{t+1} - \phi_t\|_2$ shrinking as the policy stabilizes.

S7. Summary

The supplementary evidence strengthens the revised manuscript in five ways. First, the experimental settings and resource model are fully specified. Second, MAML-Select shows consistent reductions in modeled compute and energy while retaining full client coverage. Third, the resource reductions are supported by paired statistical tests over the shared seeds. Fourth, the sensitivity, ablation, and scaling studies show that the learned selector is tunable, stable under state-feature perturbations, and inexpensive to execute. Fifth, the tier-level participation analysis shows that MAML-Select keeps every device tier represented while remaining cost-aware, and the convergence study confirms that the inner and outer updates satisfy the proven descent properties and that the selector and global model both converge during training.